Spam Detection Modeling Rubric

DS 4002 – Spring 2026 - Instructors: Javier Rasero; TA: Ethan Meidinger

Individual Assignment

General Description: Submit a GitHub Repository Link containing a complete model, a dataset, documentation, and results/testing of the model.

Why am I Doing This?
This case study will produce a lightweight, applicable model that can be used to detect spam or ham messages and help guide further research into more embedded and sophisticated models. Furthermore, this project gives you relevant experience approaching a real-world problem and applying fundamental data science skills to analyze and solve the problem, as well as communicate your findings.

What am I Going to do?
Follow the directions outlined in the hook statement, and the "ReadMe" in the repository, which is also linked below. Use the supplemental resources linked in the repository if more information or guidance is needed. Start by finding an applicable dataset with messages in English and a classification as spam or ham. Use the provided dataset or find your own. Ensure the dataset is accessible and available to use with no licensing or content restrictions. Then create a model and perform relevant tests and EDA as needed before uploading all your results to a GitHub Repository. Use the rubric below for more detailed guidelines and reach out to the course instructors or TAs with any questions.

Tips for Success:
- Engage your curiosity but maintain discipline and focus to ensure all work stays relevant to the main goal of the project.
- Prioritize well-kept code, utilizing comments, headings, and multiple code cells for easy
- Trust your teammates, push them to learn new things, and engage in meaningful debate when solving problems
- Evaluate your results! Your interpretation of your model and key performance metrics is just as important as the model results itself

How will I know I have succeeded? You will have created a model with meaningful interpretations of your work and met the guidelines presented in the rubric below.

<table>
<tr><th>Spec Category</th><th>Spec Details</th></tr>
<tr><td>Formatting</td><td>One Github Repository (submitted via link on canvas)
To ensure reproducibility, the repository will adapt parts of the TIER Protocol 4.0. In a nutshell, the top level page of the repository should contain:
A README.md file (which auto displays)
A LICENSE.md file (use MIT as default)
A SCRIPTS folder
A DATA folder
AN OUTPUT folder</td></tr>
<tr><td>README.md</td><td><u>Goal:</u> Orient users to your repository’s organization and content
Section 1: Explain the project and its motivation, as well as who the target audience is. This should not get overly technical. Include what the outputs and results of the project should be.
Section 2: Outline the overall structure of your repository. You do NOT need to include every file here, rather explain what it is the “scripts” folder and the “outputs” folder.
Section 3: Include your main results here. This should be brief and easily readable at a quick glance. Evaluation metrics such as accuracies, f1 scores, and t-tests would fit well here.</td></tr>
<tr><td>LICENSE.md</td><td><u>Goal</u>: This file explains to a visitor the terms under which they may use and cite your repository.
Select an appropriate license from the GitHub options list on repository creation. Usually, the MIT license is appropriate.</td></tr>
<tr><td>SCRIPTS folder</td><td><u>Goal:</u> This folder contains all code files for your project. The code files in here should reference datasets in the “Data” Folder
Include separate files for all major tasks. At least 3 files should be included, one for data cleaning, exploratory data analysis (EDA), and modeling</td></tr>
<tr><td>DATA folder</td><td><u>Goal:</u> Ensure transparency and reproducibility by including data.
Include your original, unedited dataset. Also include any intermediate or final datasets you use for your analysis after cleaning or editing.
Data files should be clearly and explicitly labeled differently</td></tr>
<tr><td>OUTPUT folder</td><td><u>Goal:</u> Create a folder that highlights your findings
Any major outputs (tables, charts, graphs) must be included
Model accuracy and precision MUST be included</td></tr>
<tr><td>References</td><td>All references should be listed at the end of the document
Use IEEE Documentation style (link)</td></tr>
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Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from Streifer & Palmer (2020).